



## sebastian/diff

Diff implementation for PHP, factored out of PHPUnit into a stand-alone component.

### Installation

You can add this library as a local, per-project dependency to your project using Composer:

```
composer require sebastian/diff
```

If you only need this library during development, for instance to run your project's test suite, then you should add it as a development-time dependency:

```
composer require --dev sebastian/diff
```

### Usage

**Generating diff** The `Differ` class can be used to generate a textual representation of the difference between two strings:

```
<?php
use SebastianBergmann\Diff\Differ;

$differ = new Differ;
print $differ->diff('foo', 'bar');
```

The code above yields the output below:

```
--- Original
+++ New
@@ @@
-foo
+bar
```

There are three output builders available in this package:

**UnifiedDiffOutputBuilder** This is default builder, which generates the output close to udiff and is used by PHPUnit.

```
<?php

use SebastianBergmann\Diff\Differ;
use SebastianBergmann\Diff\Output\UnifiedDiffOutputBuilder;

$builder = new UnifiedDiffOutputBuilder(
```

```

        "--- Original\n+++ New\n", // custom header
        false                       // do not add line numbers to the diff
    );

```

```

$differ = new Differ($builder);
print $differ->diff('foo', 'bar');

```

**StrictUnifiedDiffOutputBuilder** Generates (strict) Unified diff's (unidiffs) with hunks, similar to `diff -u` and compatible with `patch` and `git apply`.

```
<?php
```

```

use SebastianBergmann\Diff\Differ;
use SebastianBergmann\Diff\Output\StrictUnifiedDiffOutputBuilder;

```

```

$builder = new StrictUnifiedDiffOutputBuilder([
    'collapseRanges'    => true, // ranges of length one are rendered with the trailing `
    'commonLineThreshold' => 6,  // number of same lines before ending a new hunk and creating a new one
    'contextLines'      => 3,    // like `diff: -u, -U NUM, --unified[=NUM]`, for patch/apply
    'fromFile'          => '',
    'fromFileDate'      => null,
    'toFile'            => '',
    'toFileDate'        => null,
]);

```

```

$differ = new Differ($builder);
print $differ->diff('foo', 'bar');

```

**DiffOnlyOutputBuilder** Output only the lines that differ.

```
<?php
```

```

use SebastianBergmann\Diff\Differ;
use SebastianBergmann\Diff\Output\DiffOnlyOutputBuilder;

```

```

$builder = new DiffOnlyOutputBuilder(
    "--- Original\n+++ New\n"
);

```

```

$differ = new Differ($builder);
print $differ->diff('foo', 'bar');

```

**DiffOutputBuilderInterface** You can pass any output builder to the `Differ` class as long as it implements the `DiffOutputBuilderInterface`.

**Parsing diff** The Parser class can be used to parse a unified diff into an object graph:

```
use SebastianBergmann\Diff\Parser;
use SebastianBergmann\Git;

$git = new Git('/usr/local/src/money');

$diff = $git->getDiff(
    '948a1a07768d8edd10dcefa8315c1cbefb31833',
    'c07a373d2399f3e686234c4f7f088d635eb9641b'
);

$parser = new Parser;

print_r($parser->parse($diff));
```

The code above yields the output below:

```
Array
(
    [0] => SebastianBergmann\Diff\Diff Object
        (
            [from:SebastianBergmann\Diff\Diff:private] => a/tests/MoneyTest.php
            [to:SebastianBergmann\Diff\Diff:private] => b/tests/MoneyTest.php
            [chunks:SebastianBergmann\Diff\Diff:private] => Array
                (
                    [0] => SebastianBergmann\Diff\Chunk Object
                        (
                            [start:SebastianBergmann\Diff\Chunk:private] => 87
                            [startRange:SebastianBergmann\Diff\Chunk:private] => 7
                            [end:SebastianBergmann\Diff\Chunk:private] => 87
                            [endRange:SebastianBergmann\Diff\Chunk:private] => 7
                            [lines:SebastianBergmann\Diff\Chunk:private] => Array
                                (
                                    [0] => SebastianBergmann\Diff\Line Object
                                        (
                                            [type:SebastianBergmann\Diff\Line:private] => 3
                                            [content:SebastianBergmann\Diff\Line:private] =>
                                        )
                                    [1] => SebastianBergmann\Diff\Line Object
                                        (
                                            [type:SebastianBergmann\Diff\Line:private] => 3
                                            [content:SebastianBergmann\Diff\Line:private] =>
                                        )
                                )
                        )
                )
        )
)
```

